SABZ
AI-ADAPTIVE TRAFFIC SIGNALS

THE PROBLEM

Traffic lights that cannot see

- Lahore moves 4.1 million vehicles through signals that follow a clock, not the traffic.
- The average commute takes 34.8 minutes, much of it spent idling at red lights.
- Fixed timers run blind cycles: an empty road can hold green while a long queue burns time and fuel on the cross street.
- The signal never sees which road actually needs the green.

THE BLIND CYCLE

NS GREEN · 30s

EW GREEN · 30s

Then repeat. The same split at rush hour and at 3 AM, whatever the road looks like.

4.1M
vehicles in Lahore

34.8 min
average commute

Result: green time is wasted where nobody is waiting, and missing where queues are longest.

THE SOLUTION

SABZ: signals that see, think, and adapt

SEE

Cameras count vehicles on every approach, in real time. No new roadworks: SABZ reads the cameras an intersection already has.

THINK

A trained DQN brain reads an 18-feature state of the junction and decides, every second, which road gets green.

ACT

Signals hand green to the road that needs it, then re-check a second later. One road open at a time.

HARD-CODED SAFETY, NEVER LEARNED AWAY

Green is never cut under a car inside the 70 m dilemma zone. A starvation guard means no road waits forever. Max-green keeps every split fair.

5,506	1 s	2 of 3	7.3 s
model parameters	decision cycle	benchmark scenarios won	best average delay

System Architecture



Prototype status: full loop proven in simulation; vehicle detection verified on recorded footage, offline.

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LIVE DEMO · RESULTS

Watch it beat the fixed timer, live

Two identical intersections. Same rush-hour traffic, same start. One runs a blind 30-second fixed timer. The other runs the real trained SABZ brain, deciding every second, one road open at a time.

- Identical traffic generated live on both sides
- Real trained weights: the same 5,506-parameter brain from our benchmarks
- Every decision explained on screen, in English and Urdu

2/3
scenarios won against actuated control

7.3 s
best average delay per vehicle

47%
less delay in the best live battle

SCAN FOR THE LIVE DEMO

Run the battle on your phone: fixed timer vs the SABZ brain.

github.com/pakmultilinks-dot/sabz